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# Half-lives and the random nature of radioactive decay

**Atomic structure**
Paper 1 · Combined and Separate
8 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 2 starred (★) questions first.

| | QUESTION | ANSWER | ✓ |
|---|---|---|---|
| 1 | **Define half-life in terms of activity.** | The time taken for the activity (or count rate) of a sample to halve. | ☐ |
| 2 ★ | **Define half-life in terms of the number of nuclei.** | The time taken for the number of nuclei of the isotope in a sample to halve. | ☐ |
| 3 | **Describe what happens to the half-life of an isotope as time passes.** | Nothing: it is constant. | ☐ |
| 4 | **Describe what happens to the activity of a radioactive sample over time.** | It decreases. | ☐ |
| 5 | **Describe what happens to the number of unstable nuclei in a radioactive sample over time.** | It decreases. | ☐ |
| 6 | **State the fraction of radioactive nuclei remaining after two half-lives.** | One quarter (¼) | ☐ |
| 7 | **State the fraction of radioactive nuclei remaining after three half-lives.** | One eighth (⅛) | ☐ |
| 8 ★ | **Describe how to find the half-life from a graph of activity against time.** | Find the time taken for the activity to fall to half its starting value. | ☐ |

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